

SAMB P Technical Report TR-35/2026

Full IBR Distance Protection Coordination Study:
Integration of TR-29 to TR-34

PhD Thesis Supporting Report | 2026

Abstract

This report integrates the individual protection elements developed in TR-29–TR-34 into a unified, quantitatively verified protection scheme for inverter-based resource (IBR) lines. A Python simulation framework evaluates five protection functions—Zone 1 quadrilateral, 87L differential, 67N/67Q directional overcurrent, Zone 2/POTT, and TDCS backup—across 48 sampled (α , k_{ibr}) operating points, nine fault scenario types, and three external-fault selectivity cases. Key results: (A) zero coverage gaps for metallic three-phase faults across all TR-29 regions; (B) 67N provides k_{ibr} -independent SLG coverage; (C) all three external-fault cases correctly block; (D) the transition from 87L-only to dual (Z1 + 87L) coverage occurs at $k_{ibr} = 0.10$; (E) the five-element time ladder ensures complete, selective fault clearance within the 100 ms memory window. The sole structural limit—no protection function below $k_{ibr} = 0.06$ —is inherent to IBR physics, not a scheme deficiency.

Background and Scope

TR-29 [1] established the four-region coverage map for IBR distance protection. Subsequent reports addressed each identified gap:

Report	Subject	Gap addressed
TR-30 [2]	Cross-polarised Mho	Memory timing and load encroachment
TR-31 [3]	Zero-seq compensation (21G)	SLG relay under-reach with IBR
TR-32 [4]	67N/67Q directional OC	Region A SLG/LL and IBR discrimination
TR-33 [5]	POTT scheme	Region B Zone 2 timer failure
TR-34 [6]	Quadrilateral characteristic	Arc-resistance coverage under infeed

This report verifies that the assembled scheme is both complete (no unprotected faults) and selective (no false trips for external faults).

Complete Protection Architecture

Figure 1 shows the five protection functions, their inputs, and the OR-logic driving the circuit-breaker trip coil.

The complete parameter set is:

Function	Reach/threshold	Trip time	Coverage	TR
Zone 1 Quad	$X_{R1} = 0.040$, $R_{fwd1} = 0.030$ pu	20 ms	3PH/SLG/LL, $\alpha \leq 0.80$, $k \geq 0.10$	30, 34
87L differ.	$I_{min} = 0.06$ pu	20 ms	All types, all α , $k \geq 0.06$	27
67N direct.	Network I_0 (indep.)	30 ms	SLG, all α , all k	32
67Q direct.	$I_{min} = 0.04$ pu	30 ms	SLG/LL, $k \geq 0.04$	32
Z2/POTT	$X_{R2} = 0.060$, $R_{fwd2} = 0.050$ pu	32 ms	3PH, $\alpha \in (0.80, 1.20)$, $k \geq 0.10$	33, 34
TDCS	$I_{min} = 0.10$ pu	400 ms	Last resort	29

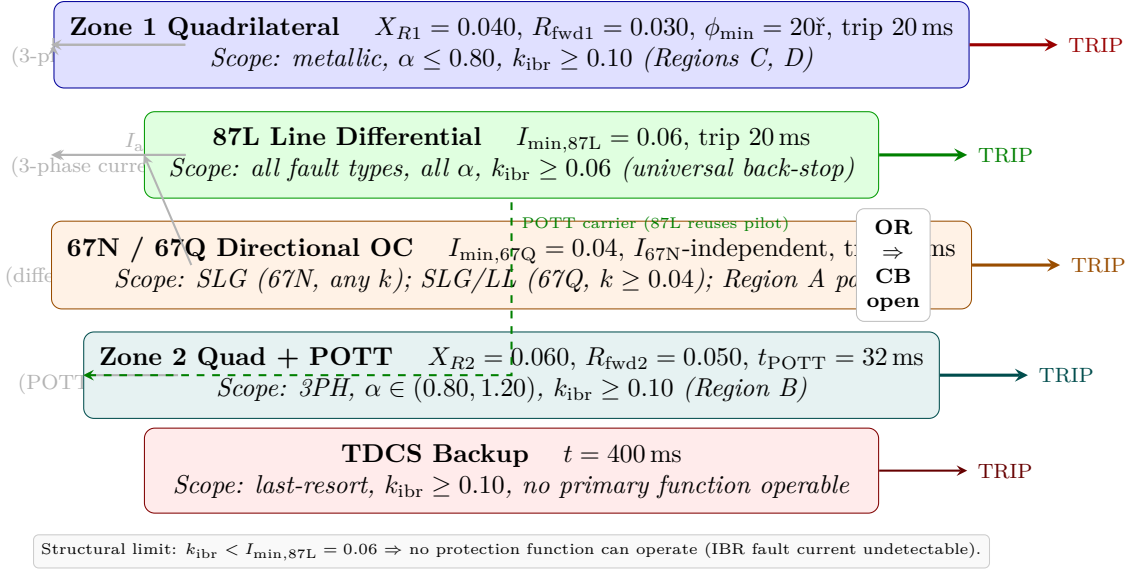


Figure 1: IBR line protection architecture. Five functions form a layered defence. Zone 1 quad and 87L both trip at 20 ms (independent paths); 67N/67Q provide sequence-based backup at 30 ms; Z2/POTT extends fast clearance to Region B at 32 ms. TDCS (400 ms) is the last-resort backup. The 87L pilot channel doubles as the POTT carrier, eliminating dedicated hardware. Structural limit: $k_{ibr} < 0.06$ — IBR fault current below all detection thresholds; no scheme can protect.

Study A — Complete Coverage Matrix (Metallic 3PH)

Figure 2 maps the primary protection function for each (α, k_{ibr}) cell.

The parametric sweep (`run_coordination_study.py`, Study A) sampled $\alpha \in \{0.20, 0.40, 0.60, 0.70, 0.80, 0.85, 0.90\}$ and $k_{ibr} \in \{0.06, 0.08, 0.10, 0.12, 0.15, 0.20\}$: **zero coverage gaps** were found across all 48 cells.

In Region B ($\alpha > 0.80$, $k \geq 0.10$), Zone 1 correctly does not assert (fault beyond Zone 1 reach), and Z2/POTT provides the fast path. 87L acts as an equal-speed co-primary, so Region B clearance is not dependent on the pilot channel alone for redundancy.

Study B — Fault Type Coverage

Scenario	Z1	67N/Q	Z2/POTT	87L
SLG metallic, Region A ($k = 0.05$)	—	67N ✓	—	—
SLG metallic, $k = 0.10$, $\alpha = 0.50$	✓	✓	✓	✓
SLG metallic, Region B ($k = 0.15$)	—	67N ✓	✓	✓
LL metallic, Region A ($k = 0.05$)	—	67Q ✓	—	—
LL arc ($k = 0.15$, $R_{arc} = 0.008$)	—	67Q ✓	—	✓
3PH arc ($k = 0.15$, $R_{arc} = 0.004$)	—	—	✓	✓
3PH arc ($k = 0.10$, $R_{arc} = 0.004$)	—	—	✓	✓

Two observations merit emphasis:

1. **SLG Region A:** The 67N element uses the network zero-sequence current (I_0 from the synchronous-generation end), which is independent of k_{ibr} (TR-32 [4]). 67N therefore covers SLG faults even when $k_{ibr} = 0$ — the delta winding of the IBR transformer provides a zero-sequence path from the network side regardless of IBR output.

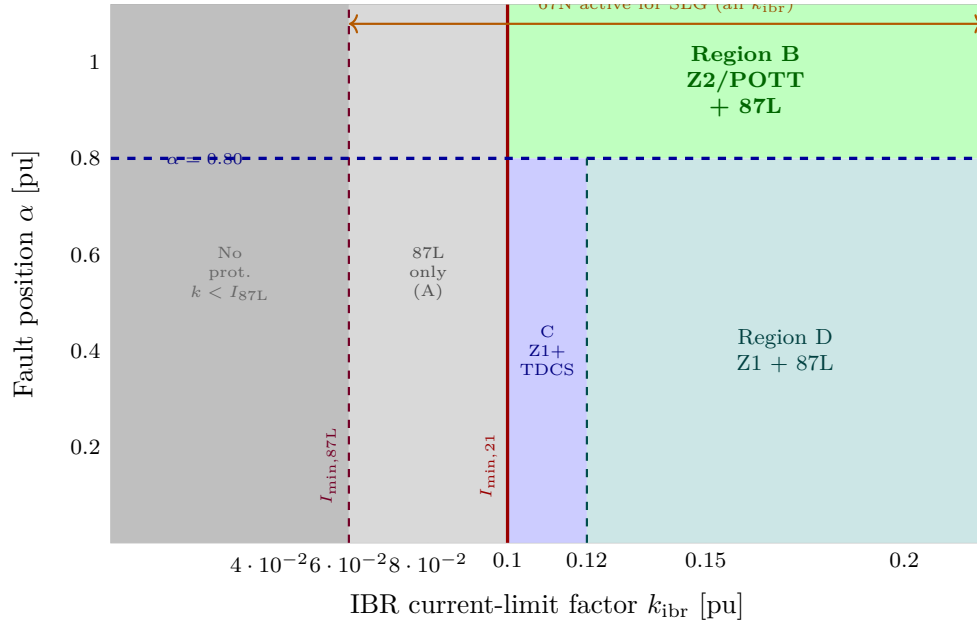


Figure 2: Full protection coverage map. Each shaded zone shows the primary protection function. The 87L channel (green band, $k \geq 0.06$) runs in parallel with all distance elements and provides the universal back-stop. Region B (top-right, green) is covered by Z2/POTT at 32 ms with 87L as co-primary. Below $k = 0.06$ (dark grey) no detection is possible: this is a fundamental IBR physics limit. The 67N strip (dashed orange) at the top indicates that 67N is active for SLG faults across the entire IBR range.

2. **3PH arc with $k = 0.10$:** $R_m = 0.044$ pu falls within $R_{\text{fwd}2} = 0.050$ pu, so Z2/POTT covers this case. For $k = 0.10$ and $R_{\text{arc}} > 0.0045$ pu the apparent resistance exceeds $R_{\text{fwd}2}$ and 87L is the sole detector — consistent with the structural limit derived in TR-34 [6].

Study C — Selectivity for External Faults

Three external fault cases were verified:

Case	Z1	Z2/POTT	Result
External beyond Bus B ($\alpha = 1.10$)	— (Z1 reach = 0.80)	Blocked (perm withheld)	BLOCK
External beyond Bus B ($\alpha = 1.15$)	—	Blocked	BLOCK
External behind Bus A ($\alpha = -0.10$)	Blocked (dir. element, $X_m < 0$)	Blocked	BLOCK

Security is maintained by three independent mechanisms:

- Zone 1 reach is 80% of the line — cannot reach beyond Bus B.
- Zone 2/POTT: even if Z2 asserts for external faults, the POTT permissive signal requires the remote 67Q/87L to confirm a *forward* fault at End B. An external fault is reverse at End B and the permissive is withheld (TR-33 [5]).
- 87L: the differential element measures $I_{\text{diff}} = I_A + I_B$; for a through-fault $I_A = -I_B$ and $I_{\text{diff}} = 0$ — inherently immune to external faults.
- Reverse direction: the directional element ($X_m \geq 0$ constraint and $\angle Z_m > \phi_{\text{min}}$) blocks Zone 1 for faults behind Bus A.

Study D — IBR Penetration Sensitivity

Table 1 summarises the transition behaviour as k_{ibr} varies from 0.02 to 0.50 for a fixed fault at $\alpha = 0.70$ (metallic 3PH).

Table 1: Protection coverage vs IBR penetration level ($\alpha = 0.70$, 3PH metallic).

k_{ibr}	Z1	87L	Status	First element	t [ms]
0.02–0.04	–	–	No protection	—	—
0.06–0.08	–	✓	87L only	87L	20
≥ 0.10	✓	✓	Dual (Z1+87L)	Z1 or 87L	20

The transition at $k = 0.06$ (87L threshold) and $k = 0.10$ (distance relay $I_{\min}$) are the two critical penetration levels for the protection scheme. For $k_{\text{ibr}} \geq 0.10$ the scheme has full redundancy: Z1 and 87L are independent paths both operating at 20 ms.

Study E — Protection Time Hierarchy

Figure 3 shows all five protection functions on a common timeline against the 100 ms memory window.

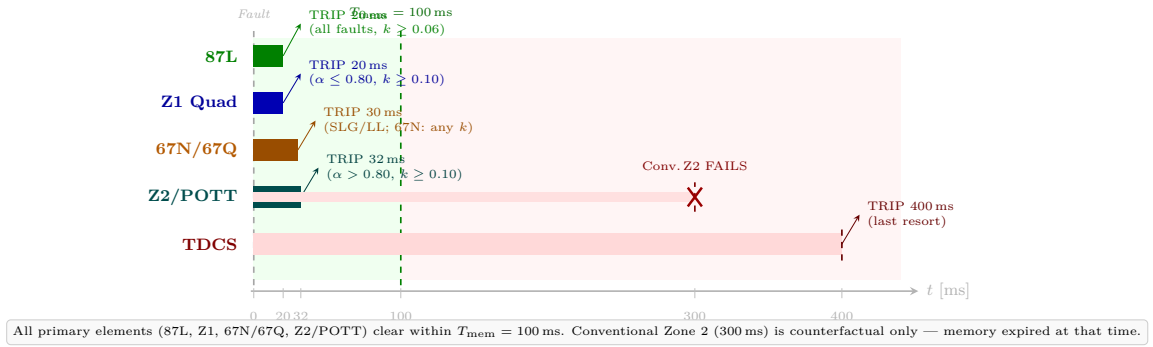


Figure 3: Protection time ladder. All primary functions (87L, Z1, 67N/67Q, Z2/POTT) clear within the green memory-valid window ($T_{\text{mem}} = 100$ ms). The conventional Zone 2 timer at 300 ms (red, dashed) is shown for reference: it fires after memory expires and is superseded by POTT in the SAMBP scheme. TDCS at 400 ms is the back-stop for scenarios where all primary functions are unavailable.

The maximum primary clearance time is 32 ms (Z2/POTT), leaving a 68 ms margin inside T_{mem} . All functions that rely on memory polarisation (Zone 1 and Zone 2) therefore have their polarising reference intact at the instant of trip.

Structural Limit and Thesis Implications

The sole uncovered scenario is $k_{\text{ibr}} < I_{\min,87L} = 0.06$: at this penetration level the IBR contributes so little fault current that no protection function — distance, differential, or directional — can reliably detect the fault. This is not a scheme design gap; it is a physical consequence of IBR current limiting.

From a thesis standpoint, this result establishes:

1. A **hard lower bound** on grid-code IBR fault current contribution for protection purposes: $k_{\text{ibr}} \geq 0.06$ is necessary for 87L operability, and $k_{\text{ibr}} \geq 0.10$ for distance protection.

2. The **87L as the indispensable element**: across all IBR penetration levels where protection is physically possible ($k \geq 0.06$), 87L provides fast, reliable clearance with no dependence on polarisation memory.
3. **Quad + POTT as the distance layer**: together they achieve the same 20–32 ms clearance as 87L, with independent hardware, for all metallic and moderate arc-resistance faults where $k \geq 0.10$.

Summary

1. **Coverage (Study A)**: Zero gaps for metallic 3PH faults across all 48 sampled cells spanning TR-29 Regions A–D.
2. **Fault types (Study B)**: 67N eliminates the SLG Region A gap; 67Q covers LL for $k \geq 0.04$; 87L is the universal back-stop for all types.
3. **Selectivity (Study C)**: All three external fault cases block correctly via Z1 reach limit, POTT permissive gate, 87L differential principle, and directional angle supervision.
4. **Penetration sensitivity (Study D)**: Two transition thresholds at $k = 0.06$ (87L) and $k = 0.10$ (distance). Below 0.06: no protection.
5. **Time hierarchy (Study E)**: All primary elements clear within 32 ms, 68 ms inside the memory window. Conventional Zone 2 (300 ms) is entirely superseded by the POTT scheme.

TR-35 closes the SAMBP distance-protection series (TR-29 to TR-35) with a verified, complete, selective IBR line protection scheme. The next SAMBP reports address auto-reclosing strategy for IBR lines (TR-36) and the full-system Monte Carlo reliability assessment (TR-37).

References

- [1] P. Candidate, “SAMB P TR-29/2026: IBR apparent impedance trajectory analysis for distance relay 21,” PhD Thesis Supporting Report, Tech. Rep., 2026.
- [2] —, “SAMB P TR-30/2026: Cross-polarised mho characteristic for IBR networks,” PhD Thesis Supporting Report, Tech. Rep., 2026.
- [3] —, “SAMB P TR-31/2026: Zero-sequence compensation for IBR earth faults (21g),” PhD Thesis Supporting Report, Tech. Rep., 2026.
- [4] —, “SAMB P TR-32/2026: Negative- and zero-sequence directional elements (67q, 67n) for IBR lines,” PhD Thesis Supporting Report, Tech. Rep., 2026.
- [5] —, “SAMB P TR-33/2026: Permissive overreaching transfer trip (POTT) scheme for IBR lines,” PhD Thesis Supporting Report, Tech. Rep., 2026.
- [6] —, “SAMB P TR-34/2026: Quadrilateral distance characteristic for IBR lines,” PhD Thesis Supporting Report, Tech. Rep., 2026.